Higher National Diploma in Quantity Surveying
2nd Year, 1st Semester Examination- 2022
QS 21013 – Measurement II

Instructions for Candidates:	No. of questions : 07
Answer 05 questions only	No. of pages : 07
All questions carry equal marks	Time : 03 hours

- Clean unmarked copy of standard Method of Measurement of Building Works (SLS 573) is permitted to use in the exam.
- You may use the below given equation when calculating reinforcement requirement Unit weight of steel (kg/m) = (Diameter of steel in mm)^2/162.162
- All drawings are not to scale.

01.

There are two different kinds of separate footings in a building structure: type C1 (06 Nos.) and type C2 (05 Nos.) Take-off quantities will be calculated utilizing the data from Dwg Nos 1,2,3,4,5 and Table 2,3. (Concrete grade M25; specifics of the clear cover are provided). Any assumptions you made ought to be made explicit. Takeoff dimensions for the following work items.

Assume that the plan has beams only on its horizontal edges.

a. Excavation for column footing (03 Marks)
b. Screed concrete for column base (03 Marks)
c. Concrete for columns and column footing (03 Marks)
d. Formwork for sides of foundation (04 Marks)
e. Formwork for soffit of slabs (03 Marks)
f. Formwork for edges of beam (04 Marks)

(Total 20 Marks)

02. You have to takeoff measurements for Reinforcement work for following work items by using the data from DWGs 1 and 2. You should prepare a bar schedule for all work items and clearly indicate the formulas of length calculation as well.

a) Reinforcement work for Column Base (04 Marks)
b) Reinforcement work for columns up to Ground Level (GL) (04 Marks)
c) RF work columns above GL (04 Marks)
d) Reinforcement work for Beams at Slab level (04 Marks)
e) Prepare a BOQ for works described at question 1 and 2 (04 Marks)

(Total 20 Marks)

03. Takeoff measurements for following work items by using DWGs 4 and 5

a. Floor finishes for Bed room 1 and 2 (04 Marks)
b. Wall finishes for Bathroom height up to 1.5m (04 Marks)
c. Brick work for external walls (04 Marks)
d. Painting work for internal walls (04 Marks)
e. Plastering work for internal and external walls (04 Marks)

(Total 20 Marks)

04. Takeoff measurements for following work items by using Dwg Nos 6 and 7.

a. Wall plates 200 x 50mm (05 Marks)
b. Ceiling joists 50 x100mm at 400mm center to center (05 Marks)
c. Ridge and Hip rafters 50 x225 mm (05 Marks)
d. Pitch roof covered with corrugated asbestos roof covering (05 Marks)

(Total 20 Marks)

05.

a. Give brief explanation of role of QS after award of the contract? (05 Marks)
b. Identify the elements of the interim payment application. (05 Marks)
c. Give a few examples of the conditions of contract used in the Sri Lankan construction sector. (05 Marks)
d. Give brief explanation about followings, using examples (05 Marks)
   i. Preliminaries
   ii. Provisional Sum

(Total 20 Marks)

06.

a. Takeoff quantities for followings by using the data from Dwg Nos 6,7,8 and Tables 4,5.

i. Eaves soffit (04 Marks)

ii. Valance Board (04 Marks)

iii. Rafters 50x 225 mm sized (04 Marks)

b. Explain the meaning of "method of measurement" and provide examples. (04 Marks)

c. Give a brief definition about "Bills of Quantities" in construction (04 Marks)

(Total 20 Marks)

07.

a. Write down any 02 measurement rules related to measurement of structural steel in a roof. (04 Marks)

b. What is the meaning of Prime Cost Sum (PC Sum)? (02 Marks)

c. Briefly explain the steps in making a Boq. (04 Marks)

d. Draw and name the terminology of a 2-flight stair case. (04 Marks)

e. Discuss the following Topics. (06 Marks)

i) Nominated Sub Contractors

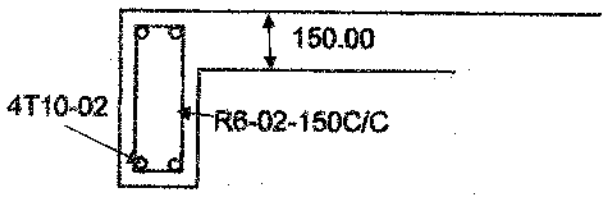
ii) Conditions of Contract

iii) Reinforcement vs. Structural steel

(Total 20 Marks)

Table 1: Door and window Schedule

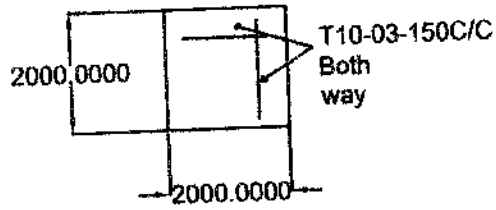
Type	Description	Size (mm)
D1	Panel door with door frame 100 x 75 mm	1200 x 2100
D2	Panel door with door frame 100 x 75 mm	1100 x 2100
W1	Glazed window with louver at the top	1200 x 1200
W2	Glazed window with louver at the top	1600 x 1200



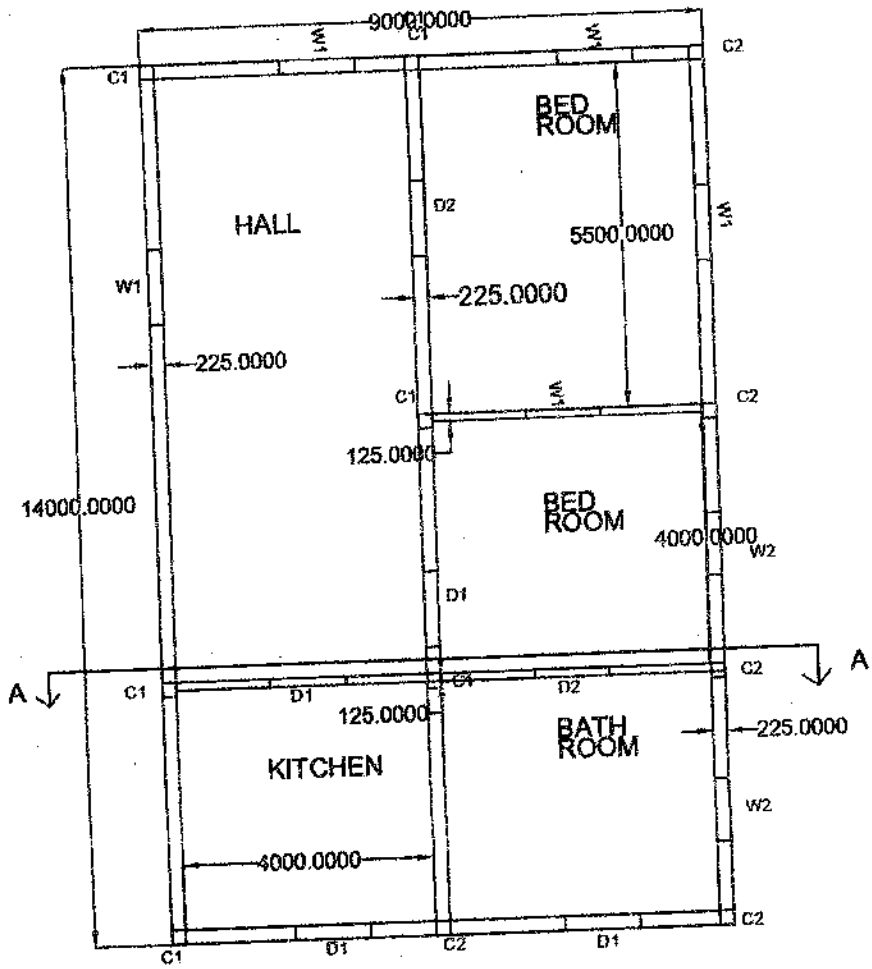
Dwg No 01: Beam section
Beam size 300 x 400 mm

COLUMN SECTION

Dwg No 02; Column section
Column size 300 x 300 mm



Dwg No 03; Column footing plan



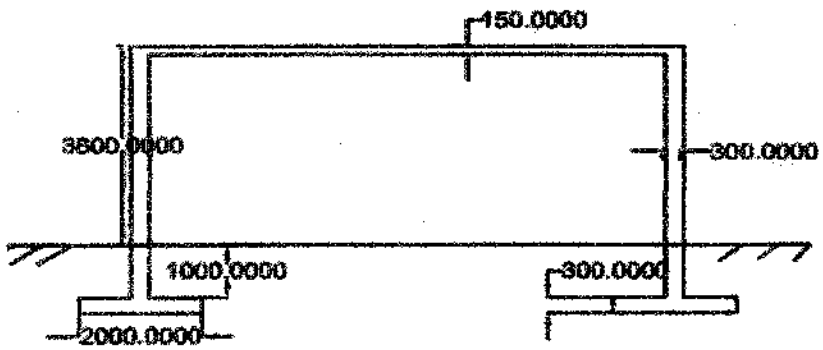
Dwg No 04: Building plan

Table 2: Details of Column footing

Column Type	Footing Size	Column Size
C1	2000 x 2000	300 x 300
C2	1800 x 1800	300 x 300

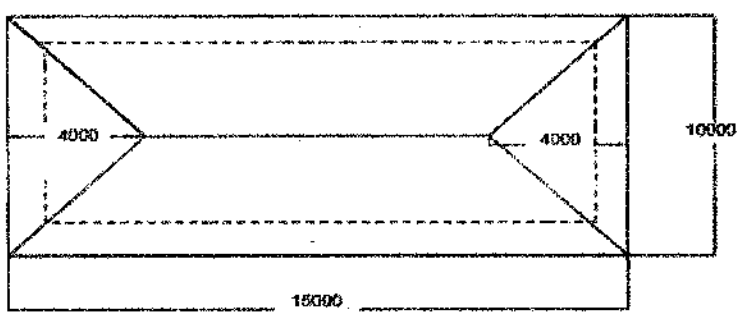
Table 3: Details of clear cover

Element	Clear Cover
Column footing	50 mm
Columns	30 mm
Beams	40 mm

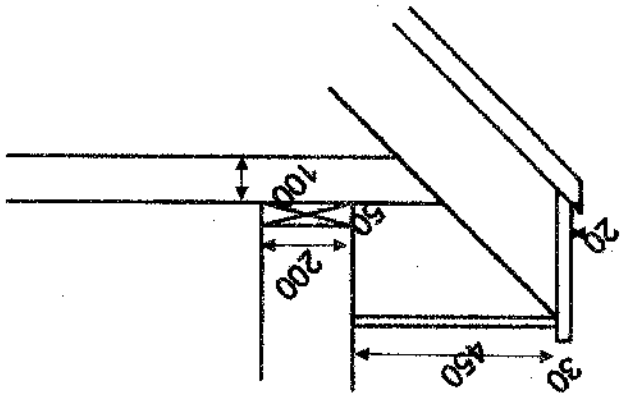


Beam B1

Dwg No 05; Building cross section



Dwg No 06; roof plan



Dwg No 07; roof cross section